

Video Transcript

Video 7: Initializing Databases (03:51)

Some databases like Mongo and Redis are ready to go out-of-the-box. You create the container and you're ready to go. You can start writing to them. But others require more structure. Relational database management systems require you to create a database, to create a table before you can start writing to them. And because of that, you might want to have a block in your automation script to be able to initialize databases. And so that is precisely what we will do. We will do it for my SQL. And as you can see here, I have a file ready to go. I'm going to start off by importing the OS and system packages. I'm then going to define two arguments as you can see there. One for create to create the containers and another to init. And I'm noting there once again that for certain databases you need to init, for others it is not needed. The very first block, the very first function is going to create the container. The next block is going to do the init. On my SQL, you can see there that I am connecting to the container. I'm then going to create a cursor, make sure that database doesn't exist before I delete it. In this case, it wouldn't exist, but I'm just doing defensive programming and then I go through and I create a table and then simply clean up by closing my connections and committing my changes.

Later on, we're going to go ahead and read that argument from the command line to be able to pick it up and then decide which block of code below that is the one that we want to run. The very first one is looking at the case of creating the database, and the second one is looking at init. So, let's go ahead and save. And as you can see here, I have my Docker dashboard. So, we can see when the container is created. And we're going to start off and run the first option, which is the option to create. And so we're going to go ahead and enter Python. And then the name of the file is containers. The option or the argument we're going to pass in is the one of create. We're going to go ahead and hit return. And as you can see there, we have created a container that's up and running, nice and green. And now, we can go ahead and run the init. As you can see there, we get an error. And that is because, as you can see here, I have not added the driver to be able to connect to MySQL. So, let's go ahead and add that.

I'm going to go ahead and save. Let's go ahead and run once again. And it looks like I left some

code that doesn't belong in this trimmer version. So, let's go ahead and scroll down. And as you can see, yes, I did leave Cassandra there. I will leave that for you to do as an exercise. But for now, we don't need it. We don't have the code that is going to perform that init. I'm going to go ahead and save once again. And now, I'm going to go ahead and run one more. And as you can see there, that went through without any errors. The database now has that new database and that new table. So, now it's your turn. Make sure you can replicate the code that we just went through. Make sure that you can initialize my SQL and take a look and add one more init for Cassandra. This is the additional one that you saw there in my code. The other two, Monger and Redis do not need it. So, go ahead and enjoy, dig in, now it's your turn.

